Paper 1 – Q1

Problem: A nighttime security camera captures images containing noise and weak object boundaries. Compare Canny, LoG, and DoG for detecting edges. Select the most suitable method and justify your choice in terms of noise handling and localization accuracy.

Selected technique: Canny Edge Detector

1. Pipeline / block diagram

PIPELINE / DIAGRAM

`Night image → Gaussian smoothing → gradient magnitude/orientation → non-maximum suppression → double threshold → hysteresis → final thin edge map`

In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Gaussian smoothing

Night-time camera images contain sensor noise and small random intensity variations. Gaussian smoothing averages nearby pixels using a distance-dependent Gaussian kernel. The standard deviation σ controls the scale of smoothing: increasing σ removes more noise but can blur small boundaries. Smoothing is therefore performed before differentiation because derivatives amplify high-frequency noise.

Output: Smoothed image with reduced random noise.

2. Gradient calculation

Horizontal and vertical derivatives Ix and Iy are calculated. Gradient magnitude identifies how rapidly intensity changes, while gradient direction gives the direction of maximum change. Object boundaries generally correspond to large gradient magnitudes.

Output: Gradient magnitude and orientation maps.

3. Non-maximum suppression

A raw gradient image often produces thick edge bands. Non-maximum suppression compares each candidate with neighbouring pixels along the gradient direction and retains only local maxima.

Output: Thin, well-localized edge pixels.

4. Double thresholding

Two thresholds classify responses as strong edges, weak candidate edges, or rejected pixels. A single threshold is less flexible because a genuine weak boundary may be below the threshold while noise may cross it.

Output: Strong and weak edge maps.

5. Hysteresis

Weak pixels are retained only when they are connected to strong edges. This preserves genuine weak boundaries while rejecting isolated noise responses.

Output: Final connected edge map.

3. Mathematical / basic principle

Principle and equations: Gaussian kernel:

$G(x,y)=1/(2\pi\sigma^2)\ \exp[-(x^2+y^2)/(2\sigma^2)]$

Gradient magnitude:

$M(x,y)=\sqrt{(Ix^2+Iy^2)}$

Gradient orientation:

$\theta=\text{atan2}(Iy,Ix)$

LoG applies the Laplacian after Gaussian smoothing:

$LoG = \nabla^2(G\sigma * I)$

DoG approximates LoG:

$DoG \approx G\sigma1*I - G\sigma2*I$

The important point is that Canny is not simply a derivative operator; it is a complete multi-stage edge detector.

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Canny	Gaussian smoothing + NMS + hysteresis	Threshold/σ tuning
LoG	Second-derivative, scale-aware	Possible double edges; scale sensitivity

DoG	Efficient LoG approximation	Less explicit edge connectivity

5. Why the selected pipeline is justified

Justification: Although LoG and DoG both include Gaussian smoothing and can locate edges through second-order behaviour or scale-space differences, Canny explicitly addresses the two requirements in the question: noise handling and localization. LoG may create double-edge responses around a boundary because a second derivative changes sign around the transition. DoG is computationally attractive and useful for scale-space processing, but it does not provide Canny's explicit non-maximum suppression plus hysteresis pipeline. Therefore the design should use Canny when thin, connected boundaries are required.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Gaussian pre-smoothing reduces noise before differentiation.
- Non-maximum suppression improves localization and produces thin edges.
- Double thresholding and hysteresis preserve weak but connected object boundaries.
- The resulting edge map is suitable for later contour extraction, Hough Transform or object detection.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- The result depends on σ and the two thresholds.
- Excessive smoothing can remove small objects or fine boundaries.
- Too low a threshold may retain noise.
- The multi-stage process is more computationally expensive than a simple Sobel operator.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: Thus, for a noisy night-time surveillance image with weak boundaries, Canny is the most suitable design because it balances smoothing, edge localization and reliable edge connectivity.

Paper 1 – Q2

Problem: A robot captures images of a warehouse containing shelves, boxes and straight structural boundaries. Design a feature-extraction pipeline using an edge detector, Hough Transform and a corner detector. Explain the purpose of each stage and the expected output after every stage.
Selected technique: Canny + Hough Line Transform + Harris Corner Detector

1. Pipeline / block diagram

PIPELINE / DIAGRAM

```
Warehouse image → Canny edges → Hough voting → detected shelf/box lines → Harris corners → fused structural feature map
```

In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Preprocessing
Convert to a suitable intensity representation and optionally smooth the image. This reduces irrelevant colour information and small sensor noise before gradients are calculated.
Output: Clean intensity image.
2. Edge detection
Canny detects localized boundaries of shelves, boxes and structural components. The output is a binary or weighted edge map rather than the final geometric interpretation.
Output: Edge pixels.
3. Hough Transform
Every edge pixel votes for possible line parameters. Peaks in parameter space identify long straight structures even when the visible edge contains gaps.
Output: Detected lines with ρ and θ parameters.

4. Corner detection
Harris examines local intensity variation in two directions. Box corners, shelf intersections and line junctions produce strong corner responses.
Output: Corner coordinates and response values.
5. Feature fusion
Lines describe global geometry while corners provide local junctions. Combining them gives a more complete representation of the warehouse layout.
Output: Structural feature set.

3. Mathematical / basic principle

Principle and equations: Canny:
$M=\sqrt{(Ix^2+Iy^2)}$

Hough line representation:
$\rho=x\cos\theta+y\sin\theta$

Harris second-moment matrix:
$M=\Sigma w[[Ix^2,IxIy],[IxIy,Iy^2]]$

Harris response:
$R=\det(M)-k(\text{trace}(M))^2$

Large positive R indicates a corner-like region.

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Canny	Localized edge map	Threshold dependent
Hough	Global straight-line geometry	Voting cost
Harris	Local junction/corner features	Scale sensitivity

5. Why the selected pipeline is justified

Justification: The techniques are complementary. Canny answers 'where are the boundaries?', Hough answers 'which boundaries form long straight structures?', and Harris answers 'where are important local junctions?'. A corner detector alone cannot describe long shelf edges, while Hough alone does not explicitly identify box corners. The sequence therefore converts pixels into progressively more meaningful geometric features. Canny is placed first because both Hough and Harris benefit from reliable local gradient information.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Hough can recover lines even when edges contain small gaps.
- Harris provides compact point features at junctions.
- The combination captures both global geometry and local structure.
- The representation is useful for robot localization, mapping and scene understanding.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Hough voting can be computationally expensive for large parameter spaces.
- Harris is sensitive to scale and smoothing.
- Canny threshold selection affects all later stages.
- Repeated textures may create many irrelevant corners or lines.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The final representation contains both line primitives and corner primitives, making it more informative for warehouse structure than either edge detection or line detection alone.

Paper 1 – Q3

Problem: A mobile application has to recognize the same product package when it appears at different sizes, rotations, and viewing angles. Compare Harris and Hessian-Affine for detecting interest points and SIFT/SURF and HOG/GLOH for describing them. Select a suitable combination and explain how scale-space or image pyramids can improve recognition.
Selected technique: Hessian-Affine detector + SIFT descriptor + Gaussian scale-space/image pyramid

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`Package image → scale-space → Hessian-Affine keypoints → scale/orientation/affine normalization → SIFT descriptor → descriptor matching → recognition`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Scale-space construction
The same image is represented after Gaussian smoothing at several σ values. A feature that is too small at one scale may become stable at another.
Output: Multi-scale image representation.
2. Hessian-Affine detection
Hessian responses identify distinctive structured regions and affine adaptation estimates an elliptical support region. This is useful when the package changes size or undergoes moderate viewpoint deformation.
Output: Repeatable affine-aware regions.
3. Orientation and region normalization
The local region is represented relative to its scale and dominant orientation. This reduces sensitivity to rotation and moderate geometric change.
Output: Canonical local patch.
4. SIFT description
SIFT computes local gradient magnitudes/orientations and forms spatial orientation histograms. The descriptor is distinctive and robust to scale and rotation.
Output: Local feature vector.
5. Matching
Query descriptors are compared with database descriptors. Distance and outlier checks are used to determine consistent package matches.
Output: Recognized package identity.

3. Mathematical / basic principle

Principle and equations: Hessian:
$H=[[I_{xx},I_{xy}],[I_{xy},I_{yy}]]$

$\det(H)=I_{xx}I_{yy}-I_{xy}^2$

Scale-space:
$L(x,y,\sigma)=G(x,y,\sigma)*I$

SIFT:
$m=\sqrt{(I_x^2+I_y^2)}$, $\theta=\operatorname{atan2}(I_y,I_x)$

SIFT descriptors aggregate gradient orientations in local cells.

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Harris	Simple corners	Not inherently scale/affine invariant
Hessian-Affine	Scale/affine adapted regions	More computation
SIFT	Distinctive scale/rotation descriptor	Heavier computation
HOG/GLOH	Shape/richer spatial gradients	Less direct for precise local matching

5. Why the selected pipeline is justified

Justification: Comparison:
Harris: simple and good for corners, but not inherently scale invariant.
Hessian-Affine: handles scale and affine-adapted regions, but costs more computation.
SIFT: strong scale/rotation robustness and distinctive local descriptor.
SURF: faster approximation to some scale-space operations but may be less distinctive in difficult matching.
HOG: strong for overall gradient/shape representation but is less natural for precise local package matching.
GLOH: richer spatial gradient layout but more complex.

Because the question explicitly includes scale, rotation and viewing-angle variation, Hessian-Affine + SIFT provides the most appropriate conceptual combination.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Scale-space prevents the detector from assuming one fixed package size.
- Affine adaptation helps under moderate viewpoint changes.
- SIFT provides distinctive local matching features.
- Local matching can work even when part of the package is not visible.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Feature detection and description are more expensive than simple corner operators.
- Large databases require efficient descriptor indexing.
- Very severe viewpoint changes can exceed the affine approximation.
- Repeated package graphics can produce ambiguous local matches.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The selected combination is Hessian-Affine for robust interest regions and SIFT for local description, supported by scale-space. The pipeline directly addresses the three variations stated in the problem.

Paper 1 – Q4

Problem: A thermal image of an industrial component contains a defective region with intensity variations and moderate noise. A few pixels inside the defective area are known. Explain how Region Growing can be used to segment the defective region. Discuss the role of seed selection, similarity threshold, and neighbourhood connectivity.
Selected technique: Seeded Region Growing with optional Gaussian/median smoothing

1. Pipeline / block diagram

PIPELINE / DIAGRAM

```
Thermal image → denoise → select defect seeds → examine neighbours → similarity test → add pixels → update region statistics → stop → defect mask
```

In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Preprocessing
Moderate noise can cause isolated pixels to satisfy or fail a similarity test incorrectly. Light smoothing can make the regional intensity distribution more stable while preserving the broad defect.
Output: Cleaner thermal image.
2. Seed selection
The known defect pixels become initial seeds. Multiple seeds should be distributed over the defect if its intensity varies significantly.
Output: Initial region R0.

3. Neighbour examination
For each current region pixel, candidate neighbours are examined. Four-connectivity uses horizontal and vertical neighbours; eight-connectivity also permits diagonals.
Output: Candidate set.
4. Similarity test
A candidate is accepted when its intensity is sufficiently close to the region's current mean, median or another robust statistic.
Output: Expanded region.
5. Region update and stopping
After adding pixels, the region statistic can be updated. Growth stops when no remaining neighbour satisfies the criterion.
Output: Final connected defect mask.

3. Mathematical / basic principle

Principle and equations: A common criterion is:
$|I(p)-\mu R| \leq T$

where I(p) is the candidate intensity, μR is the current region mean and T is the similarity threshold.

The region can alternatively use:
$\mu R-T \leq I(p) \leq \mu R+T$

Connectivity determines which neighbouring pixels are eligible.

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Region Growing	Uses known seeds/connectivity	Seed/threshold leakage
4-connectivity	Conservative	May miss diagonal continuity
8-connectivity	Captures diagonal continuity	Can join unrelated diagonal regions

5. Why the selected pipeline is justified

Justification: Seed selection is critical because Region Growing is not an automatic global clustering method; it grows from the chosen starting points. If seeds are inside the true defect, the region expands toward similar defect pixels. If seeds are placed on normal material, the method can grow the wrong region. A small T gives conservative growth and may miss part of a defect, whereas a large T allows leakage into surrounding material. Four-connectivity is conservative and avoids some diagonal connections; eight-connectivity captures diagonally connected defects but can merge regions more easily.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Simple and intuitive.
- Uses known internal defect pixels directly.
- Can represent irregular connected defects.
- Easy to implement and explain.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Sensitive to seed location.
- Threshold selection is crucial.
- Intensity inhomogeneity may break one physical defect into several regions.
- Large thresholds can cause leakage into surrounding material.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: Region Growing is appropriate because the problem explicitly provides pixels known to lie inside the defect. Good seeds, an application-specific similarity threshold and suitable connectivity are the three key design decisions.

Paper 1 – Q5

Problem: An agricultural vision system has to extract a diseased region from a leaf image containing noise, non-uniform intensity, and weak boundaries. Propose an interactive hybrid segmentation approach incorporating Region Growing, edge information, and Graph-Cut. Describe the order in which you would apply these techniques and justify the purpose of each. Explain how user-selected seeds or scribbles can guide segmentation and how the final region can be refined for subsequent disease/object detection.
Selected technique: Interactive Region Growing + Canny edge information + Graph-Cut

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`Leaf image → denoise/illumination normalization → foreground/background scribbles → Region Growing → edge map → Graph-Cut → morphological refinement → disease mask`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Preprocessing
Suppress small noise and, if necessary, reduce illumination variation. The objective is to make both region similarity and edge information more reliable.
Output: Normalized leaf image.
2. User seeds/scribbles
The user marks representative diseased pixels as foreground and healthy/background pixels as background. Several scribbles are preferable when colour/intensity varies inside the disease.
Output: Foreground/background constraints.
3. Region Growing
Starting from diseased seeds, connected pixels satisfying an appearance criterion are added. This gives a good initial estimate of the diseased region.
Output: Initial disease candidate.
4. Edge extraction
Canny or another gradient method detects strong local boundaries. These edges provide evidence about where a final segmentation boundary should be placed.
Output: Boundary evidence.
5. Graph-Cut
Construct a graph using foreground/background data costs and neighbour smoothness costs. User scribbles can be enforced as hard or strong constraints.
Output: Optimized binary segmentation.
6. Refinement
Remove isolated components, fill small holes where justified and retain the disease region. The mask can then be passed to area, shape or classification stages.
Output: Final disease mask.

3. Mathematical / basic principle

Principle and equations: Graph-Cut energy:
$E(L)=\Sigma p\ Dp(Lp)+\lambda\Sigma(p,q)Vpq(Lp,Lq)$

Dp represents how well pixel p belongs to foreground/background.
Vpq encourages spatial consistency and can be influenced by image differences.

Region Growing can use:
$|I(p)-\mu R|\leq T$

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Region/Watershed	Candidate region generation	Leakage/over-segmentation
Edge information	Explicit boundary evidence	Weak/noisy edges fragment
Graph-Cut	Global data + smoothness	Higher computation
Active Contour	Boundary refinement	Needs good initialization

5. Why the selected pipeline is justified

Justification: The order is important. Region Growing exploits regional similarity but can leak through weak boundaries. Edge information supplies explicit boundary evidence but can be fragmented by noise and weak contrast. Graph-Cut combines the two kinds of evidence globally and is constrained by user scribbles. Therefore, the hybrid is more controllable than any single method. User scribbles are particularly valuable in non-uniform leaves because they provide examples of both diseased and healthy appearance.

The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Combines regional and boundary evidence.
- Interactive seeds allow correction of ambiguous areas.
- Graph-Cut provides global optimization and leakage control.
- Final mask supports disease area, shape and classification.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Requires user interaction.
- Graph-Cut increases computation.
- Poor seeds can bias the segmentation.
- Healthy and diseased tissue with very similar appearance can remain difficult.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The proposed pipeline is a strong 20-mark answer because each stage has a distinct role: preprocessing improves reliability, seeds provide prior knowledge, Region Growing captures appearance, edges provide boundary evidence, and Graph-Cut produces the final globally consistent mask.

Paper 1 – Q6

Problem: A satellite image contains a flooded region with noise, intensity variations, and partially weak boundaries. Design a hybrid segmentation pipeline using Region Growing, an edge-based method, and Graph-Cut. Explain the sequence of these methods and the role of each stage. Discuss how seed points can be used and how the combined approach can reduce region leakage and incorrect segmentation.

Selected technique: Region Growing + Canny edge detection + Graph-Cut

1. Pipeline / block diagram

PIPELINE / DIAGRAM

`Satellite image → denoise/normalize → flood/background seeds → Region Growing → Canny edges → Graph-Cut → connected-component refinement → flood mask`

In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Preprocessing

Satellite images may contain sensor noise and illumination/atmospheric variation. Smoothing and suitable normalization improve the stability of intensity comparisons.

Output: Prepared satellite image.

2. Seed selection

Foreground seeds are placed in confidently flooded water and background seeds in clearly non-flooded land.

Output: Seed-constrained problem.

3. Region Growing

Flood seeds expand through connected pixels that resemble the current water-region appearance. This handles gradual intensity variation better than a single global threshold.

Output: Initial flood region.

4. Edge detection
Canny identifies water-land transitions where gradients are strong. These boundaries are useful for discouraging the final segmentation from crossing strong edges.
Output: Boundary map.
5. Graph-Cut
Graph-Cut combines appearance, seed and boundary terms and minimizes a global energy. Strong seed constraints prevent the algorithm from freely changing known pixels.
Output: Final flood mask.
6. Refinement
Small isolated components can be removed and the connected flood region retained. The mask can then be used for area estimation.
Output: Flood-region map.

3. Mathematical / basic principle

Principle and equations: Region criterion:
$|I(p)-\mu R| \leq T$

Canny:
$M=\sqrt{(Ix^2+Iy^2)}$

Graph-Cut:
$E(L)=\Sigma Dp(Lp)+\lambda \Sigma Vpq(Lp,Lq)$

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Region/Watershed	Candidate region generation	Leakage/over-segmentation
Edge information	Explicit boundary evidence	Weak/noisy edges fragment
Graph-Cut	Global data + smoothness	Higher computation
Active Contour	Boundary refinement	Needs good initialization

5. Why the selected pipeline is justified

Justification: Flood water is not always represented by one exact intensity because depth, sediment, shadows and imaging conditions vary. Region Growing uses local appearance and connectivity; Canny supplies boundary information; Graph-Cut combines these cues while respecting seeds. This reduces leakage into similar-looking land and avoids rejecting the whole flood region merely because some boundaries are weak.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Combines region and boundary evidence.
- Seeds provide strong application knowledge.
- Graph-Cut controls leakage.
- Useful for subsequent flood-area measurement.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Vegetation and shadows may resemble water.
- Threshold and seed placement affect Region Growing.
- Graph-Cut can be computationally demanding.
- Very weak boundaries may still be ambiguous.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The hybrid method is suitable because flood segmentation is both a regional appearance problem and a boundary problem. Combining the cues is more robust than relying on one threshold or one edge map.

Paper 1 – Q7

Problem: A manufacturing plant collects high-dimensional sensor measurements to identify abnormal machine behaviour. Only a few abnormal samples are labelled. (a) Explain how K-Medoids or GMM can be used to discover natural groups. (b) Select Bayes, KNN or ANN for abnormal-pattern identification. (c) Explain a discriminant function and the effect of outliers.
Selected technique: K-Medoids + Bayes classifier + Mahalanobis/discriminant threshold

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`Sensor vectors → scaling/PCA if needed → K-Medoids grouping → label/interpret groups → Bayes classification → discriminant score → normal/known/anomalous decision`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Preprocessing
Sensor variables such as vibration, temperature and pressure may have different units. Standardization prevents a large-unit variable from dominating distances.
Output: Normalized vectors.
2. Unsupervised grouping
K-Medoids chooses actual observations as medoids and assigns samples to the nearest medoid. It can be used to discover operating regimes without requiring labels for every observation. GMM is an alternative when soft probabilistic membership is desired.
Output: Natural operating groups.
3. Class interpretation
The few labelled abnormal samples are associated with clusters or classes. This converts discovered structure into useful machine-state information.
Output: Partially labelled groups.
4. Bayes classification
Bayes combines likelihood and prior probability. It is preferable to ANN here because only a few abnormal labels are available and a complex neural network could overfit.
Output: Posterior class probabilities/scores.
5. Discriminant
The best class score is compared with an acceptance threshold. A sample too far from every known group is treated as anomalous rather than being forced into a class.
Output: Normal/known/anomaly decision.
6. Outlier handling
Robust medoids help, but extreme observations can still affect covariance estimates or learned distributions. Outliers should be investigated rather than automatically incorporated as normal data.
Output: Robust decision process.

3. Mathematical / basic principle

Principle and equations: K-Medoids objective:
$\min \Sigma i \min j\, d(xi,mj)$

Bayes:
$P(Ck|x) \propto P(x|Ck)P(Ck)$

Mahalanobis:
$dM^2=(x-\mu)^T\Sigma^{-1}(x-\mu)$

A possible rule is:
if $\max k\, gk(x) < \tau$, flag x as anomalous.

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Selected hybrid	Complementary cues	More stages/parameters

Single detector	Simple and fast	Cannot represent all required information
Alternative robust method	May improve one property	May increase computation

5. Why the selected pipeline is justified

Justification: K-Medoids is attractive because medoids are real observations and are less sensitive to extreme values than mean-based centroids. GMM is useful when operating modes overlap and soft probabilities are desired. For the supervised stage, Bayes is appropriate when labelled data are scarce because it can work with limited examples and provides interpretable posterior scores. KNN would require representative labelled neighbours, while ANN generally benefits from more labelled examples.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Discovers hidden operating regimes.
- K-Medoids is relatively robust to outliers.
- Bayes gives probabilistic evidence.
- An explicit discriminant prevents every unknown sample from being forced into a known class.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- K-Medoids can be expensive on very large datasets.
- Bayes depends on probability-model assumptions.
- High-dimensional covariance estimation can be unstable.
- Outliers and contaminated training data can distort thresholds.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The complete approach combines unsupervised discovery with limited-label probabilistic classification and an explicit anomaly rule, matching the conditions of the manufacturing problem.

Paper 1 – Q8

Problem: A video-analysis application has to process high-dimensional visual features and then track moving objects in a fixed-camera scene. Select a suitable dimensionality-reduction technique from PCA, LDA and ICA and combine it with a non-parametric approach. Then use Background Subtraction and Optical Flow.
Selected technique: PCA + KNN + Background Subtraction + Optical Flow

1. Pipeline / block diagram

PIPELINE / DIAGRAM

```
High-dimensional features → normalization → PCA → reduced feature vectors → KNN → background model → foreground mask → optical flow → motion/tracking
```

In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Feature normalization
Different visual features can have different ranges. Standardization makes covariance and distance calculations more meaningful.
Output: Normalized feature vectors.
2. PCA reduction
PCA computes covariance eigenvectors and projects the feature vector into a lower-dimensional space containing most of the variance.
Output: Compact feature vectors.
3. KNN
The non-parametric classifier compares a new reduced vector with labelled training vectors and uses nearest neighbours. It does not require a fixed probability distribution.
Output: Class/identity decision.

4. Background Subtraction
For a fixed camera, a background model is learned. Current frames are compared with it to locate pixels belonging to moving objects.
Output: Foreground mask.
5. Optical Flow
Motion vectors are estimated between consecutive frames. The foreground mask can restrict flow analysis to relevant objects.
Output: Motion vectors.
6. Tracking
The spatial location and motion vectors are associated over time to estimate direction and movement.
Output: Object trajectory.

3. Mathematical / basic principle

Principle and equations: PCA:
$z=W^T(x-\mu)$

KNN Euclidean distance:
$d(x,xi)=\sqrt{\Sigma j(xj-xij)^2}$

Optical-flow constraint:
$Ixu+Iyv+It=0$

Flow magnitude:
$v=\sqrt{(u^2+v^2)}$

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
PCA	Fast unsupervised linear reduction	May discard low-variance discriminative information
LDA	Class-discriminative	Needs reliable labels
ICA	Independent-component representation	Not primarily variance-preserving
t-SNE	Useful nonlinear embedding/visualization	Not convenient as a direct real-time transform

5. Why the selected pipeline is justified

Justification: PCA is selected over LDA because the pipeline is not necessarily given a large set of reliable class labels for all visual dimensions; PCA is unsupervised and directly reduces correlated features. ICA is useful for source separation but is not the simplest general-purpose reduction here. KNN is a natural non-parametric method after reduction. The reduced dimension lowers the number of distance calculations, memory requirements and often the noise present in the feature vector.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Lower feature dimension reduces KNN computation.
- PCA removes redundant correlated information.
- Background subtraction works well with a fixed camera.
- Optical flow adds temporal direction and displacement information.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- PCA can remove low-variance but discriminative information.
- KNN prediction becomes costly with a huge training database.
- Shadows and illumination changes affect background subtraction.
- Optical flow is ambiguous in uniform regions.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The pipeline combines compact appearance information with explicit foreground detection and temporal motion estimation, making it suitable for a fixed-camera video system.

Paper 1 – Q9

Problem: A surveillance system extracts a large number of image features and subsequently tracks moving people. Select PCA, LDA or ICA; explain KNN on reduced vectors and explain how Background Subtraction and Optical Flow detect people and estimate direction/motion.
Selected technique: PCA + KNN + Background Subtraction + Optical Flow

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`Image features → PCA → reduced feature space → KNN → background subtraction → person mask → optical flow → direction and motion`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. PCA
The feature vector is centered and projected onto selected principal components. The number of dimensions falls from D to k, where k is much smaller than D.
Output: Reduced feature representation.
2. KNN
A query vector is compared with stored reduced vectors. The k nearest examples vote for a class. Since distance calculations now occur in a lower-dimensional space, prediction becomes cheaper.
Output: Person/object class or identity.
3. Background model
A fixed-camera system estimates the stationary scene. Pixels whose current values differ sufficiently from the background become foreground candidates.
Output: Foreground mask.
4. Optical flow
For consecutive frames, flow vectors estimate how the image pattern moves. The person mask restricts the flow statistics to relevant pixels.
Output: Foreground motion field.
5. Direction estimation
Average or median horizontal and vertical flow components provide the dominant image-plane direction. Magnitude gives relative motion speed in pixels/frame.
Output: Direction and motion estimate.

3. Mathematical / basic principle

Principle and equations: PCA:
$z=W^T(x-\mu)$

KNN:
$d=\sqrt{\Sigma(xj-xij)^2}$

Optical flow:
$Ixu+Iyv+It=0$

Image-plane motion magnitude:
$|v|=\sqrt{(u^2+v^2)}$

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
PCA	Fast unsupervised linear reduction	May discard low-variance discriminative information
LDA	Class-discriminative	Needs reliable labels
ICA	Independent-component representation	Not primarily variance-preserving
t-SNE	Useful nonlinear embedding/visualization	Not convenient as a direct real-time transform

5. Why the selected pipeline is justified

Justification: Dimensionality reduction changes KNN in two important ways. First, it reduces computational cost because each distance uses fewer dimensions. Second, it can remove correlated noise and make neighbourhoods more meaningful. However, choosing too few components can remove useful identity information. PCA is therefore preferable when an unsupervised compact representation is needed; LDA is useful when strong class labels exist, while ICA is mainly useful when independent latent sources are expected.

The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Lower-dimensional KNN is faster.
- PCA can reduce redundant/noisy dimensions.
- Background subtraction supplies object location.
- Optical flow supplies temporal direction and displacement.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- PCA is not explicitly class-discriminative.
- KNN still grows in cost with the number of training samples.
- Shadows may be classified as people.
- Camera motion and lighting changes can corrupt background modelling.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The complete solution clearly separates appearance classification from motion analysis: PCA/KNN handle compact feature recognition, while background subtraction and optical flow handle where people are and how they move.

Paper 1 – Q10

Problem: An industrial inspection system must detect scratches and textured defects whose sizes and orientations vary between images. Explain scale-space/image pyramids and Gaussian smoothing, Gabor filters and DWT, and add SIFT, SURF or HOG. Compare with only an edge or line detector.

Selected technique: Gaussian pyramid + Gabor filters + DWT + SIFT

1. Pipeline / block diagram

PIPELINE / DIAGRAM

```
Inspection image → Gaussian smoothing/pyramid → Gabor multi-orientation responses → DWT multi-resolution coefficients → SIFT descriptors → feature fusion → defect detection
```

In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Gaussian scale-space

The image is repeatedly smoothed and represented at different scales. Fine scales preserve small scratches; coarser scales make larger texture patterns easier to analyze.

Output: Multi-scale representations.

2. Gaussian smoothing

Smoothing removes high-frequency noise and makes the scale-space mathematically controlled. It must be chosen carefully because excessive smoothing can erase fine scratches.

Output: Noise-reduced images.

3. Gabor filtering

Filters with different orientations and frequencies respond to directional scratches and texture. A filter aligned with a scratch generally produces a stronger response than one perpendicular to it.

Output: Orientation/frequency response maps.

4. DWT

The image is decomposed into LL, LH, HL and HH subbands. These capture low-frequency structure and directional high-frequency

detail at multiple resolutions.
Output: Wavelet coefficient features.

5. SIFT

SIFT supplies local keypoints and gradient-orientation descriptors that are comparatively robust to scale and rotation.
Output: Local feature vectors.

6. Feature fusion

Responses from scale, orientation, frequency and local descriptors are combined before classification or detection.
Output: Hybrid defect representation.

3. Mathematical / basic principle

Principle and equations: Gaussian scale-space:
$L(x,y,\sigma)=G\sigma * I$

Gabor filtering:
Gaussian envelope × sinusoidal carrier, with orientation θ and frequency f.

DWT:
LL = approximation; LH/HL/HH = directional detail bands.

SIFT:
$m=\sqrt{(Ix^2+Iy^2)}$, $\theta=\text{atan2}(Iy,Ix)$.

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Selected hybrid	Complementary cues	More stages/parameters
Single detector	Simple and fast	Cannot represent all required information
Alternative robust method	May improve one property	May increase computation

5. Why the selected pipeline is justified

Justification: A single Canny edge detector mainly describes sharp boundaries and can miss low-contrast textured defects. Hough is excellent for explicit geometric primitives such as lines or circles but is not a general texture representation. Harris is useful for corners, whereas scratches can be elongated and corner-free. The hybrid combines scale, orientation, frequency and local gradient information, so it is better matched to defects whose size and orientation vary.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Detects features at multiple scales.
- Gabor captures orientation-specific texture.
- DWT captures localized frequency information.
- SIFT adds scale/rotation-tolerant local description.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Many filters and scales increase computation.
- Feature fusion increases dimensionality.
- Incorrect smoothing may erase fine defects.
- SIFT is more computationally demanding than a simple edge detector.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The hybrid pipeline is justified because industrial scratches are not only edges; they are textured, oriented structures that may appear at different sizes. Multi-scale and multi-frequency processing therefore gives a richer representation.

Paper 2 – Q1

Problem: Design a real-time feature extraction pipeline for pedestrian and vehicle detection in urban autonomous driving under night-time and rainy conditions. Compare Canny/Sobel/Scharr, FAST/Harris/Shi-Tomasi and ORB/BRIEF/SIFT; justify speed, robustness, repeatability and Gaussian pyramids.
Selected technique: Canny + FAST + ORB + Gaussian pyramid

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`Frame → denoise/contrast normalization → Gaussian pyramid → Canny → FAST → oriented ORB descriptor → matching/detection`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Preprocessing and pyramid
Rain, sensor noise and illumination variation are reduced using appropriate smoothing/normalization. A Gaussian pyramid creates representations at several scales so vehicles/pedestrians of different apparent sizes can be processed.
Output: Multi-scale denoised frames.
2. Canny
Canny is preferred over basic Sobel because smoothing, non-maximum suppression and hysteresis are built into the complete detector. Scharr gives improved derivative accuracy but does not provide the same full edge pipeline.
Output: Localized edges.
3. FAST
FAST tests a circle of pixels around a candidate and is computationally efficient. It is selected because autonomous driving imposes real-time constraints.
Output: Fast interest points.
4. ORB
ORB combines oriented FAST with a compact binary descriptor. Hamming distance makes matching fast.
Output: Rotation-aware binary descriptors.
5. Detection/tracking
Matched features and object-level processing produce pedestrian/vehicle candidates. Features can be associated across frames.
Output: Object candidates/tracks.

3. Mathematical / basic principle

Principle and equations: Gradient:
$M=\sqrt{(Ix^2+Iy^2)}$, $\theta=\text{atan2}(Iy,Ix)$

FAST uses a circle of neighbouring pixels and compares them with $Ip\pm t$.

ORB descriptors are binary; Hamming distance counts differing bits.

Gaussian pyramid:
$I0 \rightarrow G\sigma * I0 \downarrow 2 \rightarrow I1 \rightarrow \ldots$

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Selected hybrid	Complementary cues	More stages/parameters
Single detector	Simple and fast	Cannot represent all required information
Alternative robust method	May improve one property	May increase computation

5. Why the selected pipeline is justified

Justification: The real-time constraint makes FAST and ORB attractive. BRIEF is compact but is not inherently rotation invariant. SIFT gives stronger scale/rotation robustness but is computationally heavier. Sobel is simple and fast; Scharr improves derivative quality, while Canny adds smoothing, thinning and hysteresis. Therefore Canny + FAST + ORB provides a practical balance, while the pyramid improves scale coverage.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- High-speed binary matching.
- Canny improves edge reliability under noise.
- Pyramids support multiple object sizes.
- ORB supplies orientation information at modest cost.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Rain and headlights can create false features.
- ORB is less distinctive than SIFT in some difficult matching cases.
- Multiple scales increase computation.
- Very severe viewpoint or weather effects remain challenging.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The selected pipeline is designed around the application's most important constraint: real-time operation without abandoning basic robustness to noise, rotation and scale.

Paper 2 – Q2

Problem: Design an interactive segmentation pipeline for extracting a tumour from noisy 3-D MRI volumes with intensity inhomogeneity and weak boundaries using Active Contours, Watershed and Graph-Cut. Explain seeds, leakage prevention, topology and volume measurement.
Selected technique: Marker-controlled Watershed → Graph-Cut → 3-D Active Contour/Level Set

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`3-D MRI → denoise/bias correction → foreground/background scribbles → marker watershed → Graph-Cut → level-set refinement → tumour mask → volume`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Preprocessing
MRI may contain noise and intensity inhomogeneity. Light denoising and bias-field correction make tumour/background appearance more consistent without destroying the boundary.
Output: Prepared MRI volume.
2. User seeds
The user marks tumour voxels and definite background voxels. These are strong constraints and provide appearance examples.
Output: Foreground/background markers.
3. Marker-controlled watershed
The gradient magnitude becomes a topographic surface. Markers prevent every noise minimum from producing an independent region, reducing over-segmentation.
Output: Controlled candidate regions.
4. Graph-Cut
The graph represents voxel data terms and neighbourhood smoothness. The minimum cut separates tumour from background while respecting seeds.
Output: Globally optimized tumour mask.
5. Active contour/level set
The boundary is refined using image attraction and smoothness. A level-set surface $\phi=0$ is particularly useful for 3-D shape refinement.
Output: Smooth final boundary.
6. Volume measurement
Count tumour voxels and multiply by voxel dimensions.
Output: Tumour volume.

3. Mathematical / basic principle

Principle and equations: Graph-Cut:
$E(L)=\Sigma p\ Dp(Lp)+\lambda\Sigma(p,q)Vpq(Lp,Lq)$

Level-set boundary:
$\phi(x,y,z)=0$

Volume:
$V=N_{voxels}\,\Delta x\,\Delta y\,\Delta z$

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Region/Watershed	Candidate region generation	Leakage/over-segmentation
Edge information	Explicit boundary evidence	Weak/noisy edges fragment
Graph-Cut	Global data + smoothness	Higher computation
Active Contour	Boundary refinement	Needs good initialization

5. Why the selected pipeline is justified

Justification: Watershed alone is vulnerable to over-segmentation in noisy MRI. Graph-Cut gives global seed-constrained control and reduces leakage. Active Contours are useful for the final geometric refinement of weak/fuzzy boundaries. The sequence is therefore coarse-to-fine: markers constrain candidate regions, Graph-Cut makes the major segmentation decision, and the active contour refines the surface. This is also suitable for volume measurement because the final result is a 3-D binary mask.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Interactive control through scribbles.
- Reduced watershed over-segmentation.
- Graph-Cut combines data and boundary terms.
- Final 3-D mask directly supports volume calculation.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- 3-D graph optimization can be expensive.
- Results depend on seed quality.
- Level-set parameters affect convergence.
- Intensity inhomogeneity may still cause errors.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The pipeline satisfies the three major requirements of the question: leakage is controlled by graph constraints, over-segmentation is reduced by markers, and the final topologically coherent mask supports volume measurement.

Paper 2 – Q3

Problem: Design a hybrid anomaly-detection scheme for high-dimensional IoT sensor streams. First discover clusters using DBSCAN or Hierarchical Clustering, then apply One-Class SVM/Isolation Forest/Random Forest with an explicit decision boundary or reconstruction-error criterion. Discuss online updating.
Selected technique: DBSCAN + One-Class SVM + explicit anomaly score and validated online update

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`Sensor stream → scaling → DBSCAN density grouping → normal-region identification → One-Class SVM → anomaly score → threshold → validated model update`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Scaling and feature preparation
IoT streams can contain variables with very different ranges. Standardization makes neighbourhood density meaningful and prevents one variable from dominating.
Output: Normalized sensor vectors.
2. DBSCAN
DBSCAN groups dense regions and labels isolated points as noise. It does not require K to be known in advance and is useful for discovering normal operating modes.
Output: Clusters and noise points.
3. One-Class SVM
The classifier learns a boundary around normal observations. This is suitable when only a very small number of anomalies are labelled.
Output: Normal-support boundary.
4. Decision criterion
A sample with a score outside the learned support is flagged. A threshold can be calibrated using validation data to control false alarms.
Output: Normal/anomaly decision.
5. Online updating
The model should not immediately learn from every flagged point. Only validated normal samples should update the normal model; otherwise an attack can contaminate the boundary.
Output: Updated anomaly model.

3. Mathematical / basic principle

Principle and equations: DBSCAN uses ε and MinPts.

One-Class SVM:
$f(x)=\text{sign}(w\cdot\phi(x)-\rho)$

Anomaly rule:
flag x if $f(x)<0$
or, more generally, if $\text{anomaly_score}(x)>\tau$.

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Clustering	Discovers operating structure	High-dimensional distance/density issues
One-Class SVM	Learns normal support	Parameter/computation cost
Supervised classifier	Uses labels	Needs representative labels
Explicit threshold	Clear decision	Threshold controls false alarms

5. Why the selected pipeline is justified

Justification: DBSCAN is appropriate because the number of operating modes may not be known and isolated points can be identified as noise. One-Class SVM is preferable to a conventional supervised classifier when confirmed anomalies are very scarce. Isolation Forest is another possible choice, but the question asks for an explicit boundary or reconstruction/density criterion; One-Class SVM provides a direct boundary interpretation. Online updating must be conservative to prevent contamination.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Few anomaly labels required.
- Density-based discovery of normal regimes.
- Explicit normal-support boundary.
- Can be updated for gradual operating changes.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- DBSCAN becomes difficult in very high dimensions.
- One-Class SVM can be computationally expensive with large data.
- Poor threshold calibration causes false alarms or missed anomalies.
- Unvalidated online updates can absorb abnormal behaviour.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The proposed framework models normal behaviour first and treats deviations as suspicious, which directly addresses the scarcity of labelled anomalies in IoT streams.

Paper 2 – Q4

Problem: Design dimensionality reduction and classification for high-dimensional hyperspectral image patches followed by change detection. Incorporate PCA/LDA/t-SNE, a non-parametric classifier, Background Modelling and Optical Flow or Frame Differencing.
Selected technique: PCA + KNN + Background Modelling + Frame Differencing

1. Pipeline / block diagram

PIPELINE / DIAGRAM

`Hyperspectral patch → normalization → PCA → reduced spectral vector → KNN → background representation → frame difference/change score → change mask`

In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Spectral preprocessing

Normalize spectral bands and remove obvious noise/outliers. Hyperspectral vectors can contain hundreds of correlated bands.
Output: Clean spectral vectors.

2. PCA

PCA projects the correlated spectral vector into a small number of principal components. This reduces redundancy while preserving most variance.
Output: Compact spectral representation.

3. KNN

KNN uses nearby labelled reduced vectors to classify pixels/patches. Lower dimensionality reduces the cost of each distance calculation.
Output: Classified reduced patches.

4. Background modelling

For a temporal sequence, a reference/background representation is maintained in the reduced space. This is cheaper than modelling all original bands.
Output: Reduced-space background.

5. Change score

The current reduced vector is compared with the background using a distance or frame difference. A threshold produces a change mask.
Output: Change score and mask.

6. Optional flow

If objects move rather than simply change appearance, optical flow can be used to estimate displacement between frames.
Output: Motion/change information.

3. Mathematical / basic principle

Principle and equations: PCA:

$z=W^T(x-\mu)$

A simple reduced-space change score:

$D_t=||z_t-b_t||$

Change rule:

$D_t>\tau \rightarrow$ changed

Optical-flow constraint if used:

$Ixu+Iyv+It=0$

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
PCA	Fast unsupervised linear reduction	May discard low-variance discriminative information
LDA	Class-discriminative	Needs reliable labels
ICA	Independent-component representation	Not primarily variance-preserving
t-SNE	Useful nonlinear embedding/visualization	Not convenient as a direct real-time transform

5. Why the selected pipeline is justified

Justification: PCA is generally more suitable for near-real-time processing than t-SNE because PCA learns a direct linear projection that can be applied to new samples. LDA can preserve class separation when reliable labels are available, but the problem emphasizes high-dimensional spectral data and subsequent change detection. KNN is non-parametric and therefore does not require a specified distribution. Performing background modelling in the reduced space lowers memory and computation but requires enough components to retain change-related information.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Large reduction in storage and distance-computation cost.
- KNN is simple and non-parametric.
- Reduced-space background modelling is efficient.
- Frame differencing provides a clear change criterion.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- PCA may discard low-variance changes.
- KNN can be expensive with many stored samples.
- Background models may adapt incorrectly to slow changes.
- Threshold selection affects sensitivity.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The pipeline is suitable for near-real-time hyperspectral change detection because it compresses redundant spectral information before classification and temporal comparison.

Paper 2 – Q5

Problem: Extract roads, river banks and crop patterns from noisy multi-resolution satellite images using Image Pyramids + LoG, Gabor/steerable filters, DWT and SIFT/SURF/HOG.
Selected technique: Gaussian scale-space + LoG + Gabor/steerable filters + DWT + SIFT

1. Pipeline / block diagram

PIPELINE / DIAGRAM

```
Satellite image → Gaussian pyramid → LoG/derivatives → Gabor/steerable responses → DWT → SIFT → fused linear/texture features
```

In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Image pyramid

Generate multiple scales. Large roads and river structures may be clearer at coarse scales, while fine crop patterns require finer scales.
Output: Multi-resolution images.

2. Gaussian smoothing and LoG
Gaussian smoothing suppresses atmospheric/sensor noise; LoG detects scale-dependent intensity structures and edges.
Output: Scale-specific structural responses.
3. Gabor/steerable filters
Different orientations and frequencies represent road directions, river banks and crop rows. Steerable filters can synthesize responses at desired orientations.
Output: Orientation/frequency maps.
4. DWT
Decompose into LL, LH, HL and HH. These retain localized frequency and directional detail.
Output: Wavelet features.
5. SIFT
Detect local features and describe them using gradient orientation histograms. This adds local distinctiveness and scale/rotation robustness.
Output: Local descriptors.
6. Fusion
Combine complementary responses before classification/detection.
Output: Hybrid feature vector.

3. Mathematical / basic principle

Principle and equations: Scale-space:
$L(x,y,\sigma)=G\sigma * I$

LoG:
$\nabla^2(G\sigma * I)$

Hough line alternative:
$\rho=x\cos\theta+y\sin\theta$

DWT:
LL, LH, HL, HH subbands

SIFT:
$m=\sqrt{(Ix^2+Iy^2)}$, $\theta=\text{atan2}(Iy,Ix)$

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Selected hybrid	Complementary cues	More stages/parameters
Single detector	Simple and fast	Cannot represent all required information
Alternative robust method	May improve one property	May increase computation

5. Why the selected pipeline is justified

Justification: Canny + Hough is strong for crisp lines but is less expressive for textured crop patterns and structures that change scale. Harris emphasizes corners and can miss long linear or textured structures. The hybrid adds scale, frequency, orientation and local descriptor information. The cost is greater computation because multiple scales and filters must be evaluated.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Handles several object sizes.
- Gabor/steerable filters capture orientation.
- DWT captures localized frequency information.
- SIFT adds local scale/rotation robustness.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Large filter banks increase computation.
- Atmospheric noise can still create false responses.
- Feature fusion can create high-dimensional vectors.
- Parameter selection for scale/frequency/orientation is important.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The hybrid is appropriate because remote-sensing scenes contain both geometric and textured structures and cannot be represented adequately by only corners or straight-line edges.

Paper 2 – Q6

Problem: Compare Circular Hough vs Randomised Hough, Direct Least Squares vs RANSAC ellipse fitting, and Suzuki–Abe vs Douglas–Peucker contours for circular/elliptical traffic objects. Propose a real-time pipeline and discuss multi-resolution.
Selected technique: Canny → Suzuki–Abe contours → Douglas–Peucker simplification → Circular/Randomised Hough + RANSAC ellipse fitting → temporal verification

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`Traffic frame → illumination normalization → Canny → contours → contour simplification → geometric candidates → Hough/RHT or RANSAC ellipse → verification/tracking`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Preprocessing and edge detection
Normalize illumination where possible and use Canny to obtain localized boundaries. This reduces the number of pixels passed to geometric fitting.
Output: Candidate boundary pixels.
2. Suzuki–Abe contour tracing
Trace connected contours and preserve hierarchy. This is useful when traffic signs or covers form closed boundaries.
Output: Ordered contour points.
3. Douglas–Peucker
Simplify a contour while keeping the maximum approximation error below ε. This reduces the number of points used by later fitting.
Output: Simplified contour.
4. Circular Hough/RHT
Circular Hough votes over centre and radius; Randomised Hough samples subsets and can reduce exhaustive voting.
Output: Circle candidates.
5. Ellipse fitting
Direct Least Squares is fast but sensitive to outliers; RANSAC repeatedly fits candidate ellipses and selects a model supported by many inliers.
Output: Ellipse candidates.
6. Multi-resolution/verification
Run expensive fitting at suitable pyramid levels and verify candidates across frames or by geometric consistency.
Output: Stable object detections.

3. Mathematical / basic principle

Principle and equations: Circle equation:
$(x-a)^2+(y-b)^2=r^2$

Line Hough:
$\rho=x\cos\theta+y\sin\theta$

Douglas–Peucker:
retain a point when its perpendicular deviation exceeds ε.

RANSAC:
choose random minimal samples → fit model → count inliers → repeat → select model with maximum support.

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Circular Hough	Direct circle parameter voting	Large parameter space
Randomised Hough	Lower voting burden	Sampling risk
Direct Least Squares	Fast ellipse fit	Outlier sensitive
RANSAC	Robust to outliers/occlusion	Iterative/stochastic

5. Why the selected pipeline is justified

Justification: Comparison:
Circular Hough: accurate but can be expensive because the parameter space contains centre and radius.
Randomised Hough: reduces voting by sampling but has stochastic detection risk.
Direct Least Squares ellipse: fast deterministic fit but sensitive to outliers.
RANSAC ellipse: robust to outliers/partial contour but needs iterations and thresholds.
Suzuki–Abe: extracts connected contours/hierarchy.
Douglas–Peucker: simplifies contours and therefore reduces downstream cost.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Coarse-to-fine processing reduces the number of pixels and fitting candidates.
- RANSAC handles partial occlusion/outliers better than direct fitting.
- Contours reduce the search space before geometric fitting.
- Temporal verification reduces isolated false detections.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Hough parameter spaces can be large.
- RANSAC is stochastic and needs an inlier threshold.
- Weak illumination can break contours.
- Heavy occlusion can remove too much geometric evidence.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: A practical real-time design should avoid running expensive geometric fitting on every pixel. Edge detection and contours first reduce the search space, then robust geometric models are applied only to candidate objects.

Paper 2 – Q7

Problem: Design an interactive 2-D/3-D blood-vessel extraction pipeline from noisy angiographic CT using Region Growing, Hessian vesselness and Graph-Cut or Random Walker. Discuss seeds, leakage, thin structures and topology.
Selected technique: Hessian multi-scale vesselness → seeded Region Growing → Graph-Cut

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`Angiographic CT → denoise → Hessian vesselness at multiple scales → vessel/background seeds → Region Growing → Graph-Cut → vessel mask → centreline extraction`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Denoising
Suppress noise while retaining thin vessel boundaries. Excessive smoothing must be avoided because small vessels can disappear.
Output: Cleaner CT volume.
2. Hessian vesselness
Compute second-order derivatives and eigenvalues at multiple scales. Tubular structures show a characteristic eigenvalue pattern

and therefore receive a high vesselness response.
Output: Vesselness volume.

3. User seeds

Place foreground seeds inside reliable vessel sections and background seeds in non-vessel tissue.
Output: Seed constraints.

4. Region Growing

Grow from vessel seeds through connected high-vesselness/similar-intensity voxels. Connectivity helps preserve continuous vessels.
Output: Initial vessel mask.

5. Graph-Cut

Use vesselness/intensity as data evidence and neighbourhood smoothness to prevent uncontrolled leakage into adjacent tissue.
Output: Optimized vessel mask.

6. Centreline extraction

Skeletonize or trace the final mask. The segmentation should preserve branches and thin connections so that topology is not destroyed.
Output: Centreline/tree.

3. Mathematical / basic principle

Principle and equations: Hessian:
H=[[Ixx,Ixy,Ixz],[Ixy,Iyy,Iyz],[Ixz,Iyz,Izz]]

The eigenvalues λ1,λ2,λ3 characterize local curvature. Vesselness functions use eigenvalue ratios to favour tube-like structures.

Graph-Cut:
E=ΣD+λΣV

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Selected hybrid	Complementary cues	More stages/parameters
Single detector	Simple and fast	Cannot represent all required information
Alternative robust method	May improve one property	May increase computation

5. Why the selected pipeline is justified

Justification: Hessian vesselness must occur early because ordinary intensity Region Growing cannot reliably distinguish vessels from similarly bright neighbouring tissue. Multi-scale filtering is essential because vessels have different diameters. Region Growing then uses connectivity, while Graph-Cut provides global leakage control. User seeds provide reliable vessel/background examples. Preserving topology is important because later centreline extraction requires connected branches and should not contain artificial breaks.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Enhances thin vessels before segmentation.
- Multi-scale response handles different vessel diameters.
- Seeds provide user control.
- Graph-Cut improves leakage control and spatial consistency.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- 3-D Hessian computation is expensive.
- Vesselness scales must cover the expected vessel diameters.
- Bifurcations and crossing vessels complicate topology.
- Poor seeds can cause wrong growth.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The sequence is therefore vessel-specific and coarse-to-fine: enhance tubular structures, grow connected regions, globally correct leakage, then extract the centreline.

Paper 2 – Q8

Problem: Design a hybrid pattern-recognition scheme for high-dimensional industrial vibration data. First perform GMM/K-Means++ grouping, then Bayes/SVM/MLP, together with a Mahalanobis or residual discriminant. Discuss anomaly decision, updates, outliers and scarce labels.
Selected technique: GMM + SVM + Mahalanobis-distance discriminant

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`Vibration signal → time/frequency features → normalization/PCA → GMM operating-state grouping → SVM classifier → Mahalanobis/residual score → anomaly decision → model update`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Feature extraction
Convert vibration measurements into meaningful time-domain and frequency-domain features such as energy, spectral peaks or band statistics.
Output: Feature vector.
2. GMM grouping
GMM represents operating states as Gaussian components and gives soft membership probabilities. This is useful when machine regimes overlap.
Output: Operating-state probabilities.
3. SVM classifier
Use the limited labelled examples to learn a separating boundary for known fault classes. SVM can be effective with relatively small labelled datasets when features are well chosen.
Output: Known-class prediction.
4. Mahalanobis/residual discriminant
Measure whether the new sample is statistically consistent with the predicted class. A high Mahalanobis distance or large prediction residual indicates unusual behaviour.
Output: Anomaly score.
5. Decision and update
If classifier confidence is low or the discriminant exceeds τ, flag the sample. Model updates should use validated samples to avoid contaminating normal distributions.
Output: Final fault/anomaly decision.

3. Mathematical / basic principle

Principle and equations: GMM:
$p(x)=\Sigma k\ \pi k\ N(x|\mu k,\Sigma k)$

Mahalanobis:
$dM^2=(x-\mu)^T\Sigma^{-1}(x-\mu)$

SVM:
$w^T\phi(x)+b=0$

Example anomaly rule:
$dM^2>\tau \rightarrow$ anomalous.

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Clustering	Discovers operating structure	High-dimensional distance/density issues
One-Class SVM	Learns normal support	Parameter/computation cost
Supervised classifier	Uses labels	Needs representative labels
Explicit threshold	Clear decision	Threshold controls false alarms

5. Why the selected pipeline is justified

Justification: GMM is useful for discovering multiple operating regimes and soft memberships. K-Means++ is simpler and often faster, but GMM provides covariance and probability information that is useful for anomaly scoring. SVM is selected over MLP because the question states that labelled fault examples are scarce. The Mahalanobis score accounts for feature correlations and gives an explicit mathematical discriminant. Outliers can distort means and covariance, so robust preprocessing and validation are needed.

The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- GMM handles overlapping operating modes.
- SVM can work with limited labelled data.
- Mahalanobis distance accounts for covariance.
- Explicit thresholds make the anomaly decision transparent.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- GMM may be expensive in high dimensions.
- SVM kernel and regularization require tuning.
- Covariance matrices can be unstable with too few samples.
- Outliers can corrupt estimated distributions.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The hybrid scheme separates unsupervised operating-state discovery from supervised fault recognition and then adds a statistical novelty check, which is appropriate when labelled faults are scarce.

Paper 2 – Q9

Problem: Design dimensionality reduction and tracking for high-dimensional multi-camera surveillance features using PCA/ICA/Autoencoder, a non-parametric density estimator, Background Subtraction and Lucas–Kanade/Farneback Optical Flow.
Selected technique: PCA + KDE + Background Subtraction + Farneback Optical Flow

1. Pipeline / block diagram

PIPELINE / DIAGRAM

```
Camera features → normalization → PCA → KDE density model → background subtraction → foreground mask → dense Farneback flow → motion parameters
```

In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Feature normalization
Scale feature dimensions so the covariance structure is meaningful.
Output: Normalized feature vectors.

2. PCA
Reduce correlated features to a compact representation while retaining major variance. This reduces the dimensionality used by the density model and later storage.
Output: Reduced feature space.

3. KDE
Estimate the density of normal/typical reduced feature vectors without assuming a Gaussian distribution.
Output: Non-parametric density estimate.

4. Background subtraction
For each fixed camera, maintain a background representation and identify deviations as foreground objects.
Output: Foreground mask.

5. Farneback optical flow
Estimate dense motion vectors over the frame, then restrict or aggregate them over the foreground mask.
Output: Dense motion field.
6. Motion parameters
Compute mean/median u and v components and magnitude to estimate image-plane direction and relative speed.
Output: Object motion parameters.

3. Mathematical / basic principle

Principle and equations: PCA:
$z=W^T(x-\mu)$

KDE:
$f(x)=1/(nh^d)\Sigma K((x-xi)/h)$

Optical-flow constraint:
$Ixu+Iyv+It=0$

Motion magnitude:
$\sqrt{(u^2+v^2)}$

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
PCA	Fast unsupervised linear reduction	May discard low-variance discriminative information
LDA	Class-discriminative	Needs reliable labels
ICA	Independent-component representation	Not primarily variance-preserving
t-SNE	Useful nonlinear embedding/visualization	Not convenient as a direct real-time transform

5. Why the selected pipeline is justified

Justification: PCA is preferred when real-time processing requires a direct projection for every new vector. KDE is non-parametric and can represent complex feature distributions. Background subtraction provides spatial object localization, while Farneback provides dense motion rather than only sparse point motion. The reduced space can make background modelling cheaper, but too aggressive reduction can remove discriminative information. Camera stabilization is necessary if cameras themselves move.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Compact representation reduces downstream computation.
- KDE does not require a fixed Gaussian model.
- Background subtraction gives foreground localization.
- Dense optical flow provides rich motion information.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- KDE can struggle in high dimensions.
- Farneback is more expensive than sparse flow.
- PCA may discard useful information.
- Camera motion and illumination changes affect the background model.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The complete design balances statistical modelling and real-time motion analysis by reducing dimensions before density estimation and using foreground masks to focus optical-flow computation.

Paper 2 – Q10

Problem: A dermatological imaging system must extract textured skin lesions and oriented hair/vessel structures from multi-scale dermoscopic images with specular noise and scale variations. Integrate Scale-Space, Gabor, DWT and HOG/LBP.
Selected technique: Gaussian pyramid + Gaussian derivatives + Gabor + DWT + LBP

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`Dermoscopic image → specular/noise reduction → Gaussian pyramid/derivatives → Gabor responses → DWT → LBP → feature fusion → lesion/structure detection`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Preprocessing
Specular highlights and sensor noise can generate false texture. Suitable preprocessing should reduce these artefacts without destroying lesion boundaries.
Output: Cleaner dermoscopic image.
2. Scale-space
Gaussian pyramids represent the lesion and structures at several scales. Derivatives capture local changes and orientation information.
Output: Multi-scale derivative maps.
3. Gabor filters
Different orientations/frequencies detect hair and vessel patterns and texture inside lesions. Their localized nature makes them useful for oriented structures.
Output: Orientation/frequency maps.
4. DWT
Wavelet decomposition separates low-frequency lesion structure from localized high-frequency texture and directional detail.
Output: LL/LH/HL/HH features.
5. LBP
Local Binary Pattern captures micro-texture by comparing neighbours with the centre pixel.
Output: Local texture histogram.
6. Fusion
Combine scale, frequency, orientation and local texture features for the final classifier/detector.
Output: Hybrid descriptor.

3. Mathematical / basic principle

Principle and equations: LBP:
$LBP=\Sigma p\ s(gp-gc)2^{p}$

Scale-space:
$L=G\sigma * I$

DWT:
LL, LH, HL, HH

Gabor:
localized Gaussian envelope × sinusoidal carrier.

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Selected hybrid	Complementary cues	More stages/parameters
Single detector	Simple and fast	Cannot represent all required information
Alternative robust method	May improve one property	May increase computation

5. Why the selected pipeline is justified

Justification: Canny is useful for strong boundaries but does not fully describe texture. Harris is primarily an interest-point detector and can miss homogeneous textured regions. Hough is useful for specific geometric primitives but is not a general texture descriptor. The proposed hybrid is better because lesions contain both irregular texture and oriented structures. Scale-space handles size, Gabor handles orientation/frequency, DWT handles resolution, and LBP provides compact local texture information.

The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Strong multi-scale representation.
- Good orientation/frequency sensitivity.
- LBP is compact and effective for local texture.
- Combines complementary information.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Specular noise can disturb local texture codes.
- Many Gabor scales/orientations increase computation.
- Feature fusion increases dimensionality.
- LBP is not inherently fully rotation/scale invariant without suitable variants.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The hybrid is suitable because dermoscopic images contain heterogeneous texture and directional structures rather than only isolated edges or corners.

Paper 2 – Q11

Problem: Design a feature pipeline for lane-marking and road-sign detection in fog, rain and low light. Compare Canny/LoG/Phase Congruency; PPHT/LSD; Harris/Hessian-Affine/AGAST; SIFT/SURF/HOG. Discuss pyramids and steerable filters.
Selected technique: Phase Congruency/Canny + PPHT/LSD + AGAST + HOG + Gaussian pyramid

1. Pipeline / block diagram

PIPELINE / DIAGRAM

```
Adverse-weather frame → enhancement → Gaussian pyramid → Canny/phase-congruency edges → PPHT line segments → AGAST interest points → HOG descriptor → detection
```

In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Enhancement and pyramid
Reduce noise/contrast problems and create several scales. Lane markings may occupy different widths and signs may appear at different sizes.
Output: Multi-scale images.
2. Edge stage
Canny provides smoothing, gradient calculation, thinning and hysteresis. LoG provides second-order scale-space responses. Phase congruency is based on local phase alignment and can be useful when absolute contrast is unreliable.
Output: Edge/feature-strength maps.
3. Line stage
PPHT probabilistically samples Hough voting and is appropriate for lane-like line segments. LSD directly detects line segments and avoids a full Hough accumulator.
Output: Line segments.
4. Interest points
AGAST is selected when speed is important; Harris is classical, while Hessian-Affine offers greater scale/affine robustness at higher cost.
Output: Fast local points.
5. Descriptor
HOG summarizes gradient orientations in local cells and blocks, making it suitable for road-sign shape/appearance detection.
Output: Normalized gradient descriptor.

6. Multi-scale orientation refinement
Pyramids address size; steerable filters can efficiently represent multiple orientations.
Output: Robust multi-scale representation.

3. Mathematical / basic principle

Principle and equations: Gradient magnitude:
$M=\sqrt{(Ix^2+Iy^2)}$

Hough:
$\rho=x\cos\theta+y\sin\theta$

HOG:
local gradient orientations → cell histograms → block normalization.

Scale-space:
$L(x,y,\sigma)=G\sigma*I$

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Canny	Gaussian smoothing + NMS + hysteresis	Threshold/σ tuning
LoG	Second-derivative, scale-aware	Possible double edges; scale sensitivity
DoG	Efficient LoG approximation	Less explicit edge connectivity

5. Why the selected pipeline is justified

Justification: Adverse weather makes speed and robustness both important. Canny is a practical default because it handles noise and gives localized edges. Phase congruency can complement it when contrast changes make raw gradient magnitude unreliable. PPHT is useful for lane geometry; LSD is also efficient for line segments. AGAST and HOG are more suitable to a real-time pipeline than the heavier SIFT/Hessian-Affine combination, while pyramids add scale coverage.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Combines line geometry with shape/appearance features.
- Pyramids improve scale handling.
- AGAST supports fast feature detection.
- HOG is relatively compact and suited to shape.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Fog and rain can create false edges and line fragments.
- Phase congruency increases computation.
- HOG has less local matching invariance than SIFT.
- Parameters and weather-specific preprocessing remain important.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The combined pipeline is justified because no single detector handles line geometry, scale, adverse illumination and real-time constraints simultaneously.

Paper 2 – Q12

Problem: Design an interactive organ segmentation pipeline for noisy ultrasound volumes with fuzzy boundaries using Marker-controlled Watershed, Active Contours and Graph-Cut. Explain scribbles/seeds, over-segmentation, leakage and topology.
Selected technique: Marker-controlled Watershed + Graph-Cut + Active Contour

1. Pipeline / block diagram

PIPELINE / DIAGRAM

`Ultrasound → speckle reduction → user foreground/background scribbles → marker watershed → Graph-Cut → active contour refinement → final organ mask`

In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Speckle/noise reduction

Ultrasound contains speckle, which can produce many artificial gradient minima. Preprocessing should reduce this without over-blurring the organ boundary.

Output: Stabilized ultrasound volume.

2. Scribbles/markers

User marks definite organ and background regions. These markers provide strong constraints and appearance examples.

Output: Foreground/background markers.

3. Marker-controlled watershed

Use the gradient as a topographic surface but constrain the watershed using markers. This reduces the severe over-segmentation that unconstrained watershed can produce.

Output: Controlled regions.

4. Graph-Cut

Combine appearance terms, neighbourhood smoothness and user constraints. This helps stop leakage through weak boundaries.

Output: Globally optimized mask.

5. Active contour

Use the graph-cut result as initialization. The deformable contour evolves toward the fuzzy boundary while maintaining smoothness.

Output: Refined organ boundary.

6. Shape analysis

The final mask can be used for area, perimeter, volume and shape descriptors.

Output: Analysis-ready mask.

3. Mathematical / basic principle

Principle and equations: Graph-Cut:

$E=\Sigma D+\lambda\Sigma V$

Level-set representation:

$\phi(x,y,z)=0$

Watershed:

gradient magnitude acts as topographic height.

Active contour:

energy = image/boundary attraction + regularization.

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Region/Watershed	Candidate region generation	Leakage/over-segmentation
Edge information	Explicit boundary evidence	Weak/noisy edges fragment
Graph-Cut	Global data + smoothness	Higher computation
Active Contour	Boundary refinement	Needs good initialization

5. Why the selected pipeline is justified

Justification: The order prevents one method from doing all the work. Marker-controlled watershed provides meaningful initial regions; Graph-Cut gives global seed-constrained control and reduces leakage; active contours refine the fuzzy boundary. This is particularly important in ultrasound, where boundaries can be weak and intensity can overlap between organ and background.

The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- User interaction provides strong priors.
- Marker control reduces watershed over-segmentation.
- Graph-Cut controls leakage.
- Active contours give smooth boundaries.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Speckle remains difficult.
- Seed placement affects the result.
- Graph-Cut and active contours increase computation.
- Wrong markers can bias the entire segmentation.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The proposed coarse-to-fine sequence produces a mask suitable for later shape analysis while explicitly addressing both over-segmentation and leakage.

Paper 2 – Q13

Problem: Design a hybrid anomaly-detection framework for high-dimensional network-traffic feature vectors. Use K-Medoids/Spectral Clustering, then Label Propagation/One-Class SVM/Autoencoder and an explicit reconstruction-error or density-based discriminant. Discuss streaming updates and scarce confirmed attacks.
Selected technique: K-Medoids + One-Class SVM + density/score threshold

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`Network vectors → normalize → K-Medoids → normal traffic regimes → One-Class SVM → anomaly score → threshold → validated online update`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Feature preparation
Network traffic vectors can contain correlated and differently scaled features. Normalize and, if required, reduce dimension before clustering.
Output: Prepared traffic vectors.
2. K-Medoids grouping
Discover typical traffic regimes using representative medoids. Outlying points can be identified without requiring attack labels.
Output: Traffic clusters/noise.
3. One-Class SVM
Learn the support of normal traffic using mostly normal examples. This fits the scarce confirmed-attack condition.
Output: Normal boundary.
4. Density/score discriminant
A point outside the support or with sufficiently low normality score is declared anomalous.
Output: Anomaly score.
5. Online update
Only validated normal observations should be incorporated. Suspicious points should be quarantined from model updates.
Output: Updated normal model.

3. Mathematical / basic principle

Principle and equations: K-Medoids:
$\min \Sigma i \min j\, d(xi,mj)$

One-Class SVM:
$f(x)=\text{sign}(w\cdot\phi(x)-\rho)$

Density rule:
$p(x)<\tau \rightarrow$ anomaly

The threshold τ should be calibrated from validation/normal traffic.

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Clustering	Discovers operating structure	High-dimensional distance/density issues
One-Class SVM	Learns normal support	Parameter/computation cost
Supervised classifier	Uses labels	Needs representative labels
Explicit threshold	Clear decision	Threshold controls false alarms

5. Why the selected pipeline is justified

Justification: K-Medoids is selected over purely mean-based clustering because its representatives are actual traffic observations and are relatively robust to extreme points. One-Class SVM is suitable because attack labels are extremely scarce. Autoencoders can also model normal reconstruction error, but require more training complexity. A density or score threshold makes the final decision explicit.

The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Few attack labels required.
- Robust cluster representatives.
- Explicit anomaly score.
- Can adapt to multiple normal traffic modes.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- High-dimensional density estimation is difficult.
- One-Class SVM can be expensive for very large streams.
- Online updates risk contamination.
- Threshold choice controls false-positive rate.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The design treats abundant normal traffic as the source of the model and deviations from that learned normal structure as anomalies.

Paper 2 – Q14

Problem: Design dimensionality reduction and motion analysis for high-dimensional multi-spectral image sequences using LDA/PCA/NMF, a non-parametric method, Background Subtraction and dense Optical Flow. Discuss reduced-space background modelling and motion parameters.

Selected technique: PCA + KDE + Background Subtraction + dense Farneback Optical Flow

1. Pipeline / block diagram

PIPELINE / DIAGRAM

`Multi-spectral frame → normalization → PCA → KDE → reduced-space background model → foreground mask → dense optical flow → motion parameters`

In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Spectral normalization

Bring spectral bands to comparable scales and remove obvious corrupted measurements.

Output: Normalized spectral vectors.

2. PCA
Compress correlated spectral bands into a few components. This reduces the cost of modelling every original band.
Output: Reduced spectral representation.
3. KDE
Estimate the distribution of normal reduced features without assuming a particular parametric family.
Output: Density model.
4. Background subtraction
Maintain the expected reduced-space appearance of stationary regions and identify deviations.
Output: Foreground mask.
5. Dense optical flow
Estimate displacement vectors for many pixels, especially inside foreground regions.
Output: Dense motion field.
6. Motion parameters
Compute robust statistics of u and v, magnitude and direction; physical speed requires frame interval and scene calibration.
Output: Motion direction/speed estimate.

3. Mathematical / basic principle

Principle and equations: PCA:
$z=W^T(x-\mu)$

KDE:
$f(x)=1/(nh^d)\Sigma K((x-xi)/h)$

Flow magnitude:
$s=\sqrt{(u^2+v^2)}$

Direction:
$\theta=atan2(v,u)$

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
PCA	Fast unsupervised linear reduction	May discard low-variance discriminative information
LDA	Class-discriminative	Needs reliable labels
ICA	Independent-component representation	Not primarily variance-preserving
t-SNE	Useful nonlinear embedding/visualization	Not convenient as a direct real-time transform

5. Why the selected pipeline is justified

Justification: PCA is suitable for near-real-time spectral compression because it gives a direct projection. LDA could be preferable if strong class labels exist, while NMF is useful when non-negative parts are desired. KDE is non-parametric and flexible. Background modelling becomes cheaper in the reduced space, but the retained components must preserve the spectral differences associated with objects. Dense flow provides more complete motion information than sparse tracking points.
The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Reduced spectral computation.
- Flexible density estimation.
- Dense motion representation.
- Foreground mask can restrict expensive flow computation.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- KDE suffers in high-dimensional spaces.

- PCA can remove useful class-specific information.
- Dense flow is computationally expensive.
- Camera movement must be compensated.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The pipeline balances spectral compression and temporal analysis, making it suitable when the original multi-spectral dimensionality is too large for direct background and motion modelling.

Paper 2 – Q15

Problem: A materials-science application requires robust extraction of oriented microstructures and textured regions from multi-resolution SEM/TEM images containing significant noise and scale variations. Integrate scale-space, Gaussian derivatives, Gabor/Log-Gabor, DWT/Curvelet and SIFT/SURF/HOG.
Selected technique: Gaussian pyramid + Gaussian derivatives + Log-Gabor + Curvelet Transform + SIFT

1. Pipeline / block diagram

PIPELINE / DIAGRAM
`SEM/TEM → denoise → Gaussian scale-space → derivative responses → Log-Gabor orientation/frequency maps → Curvelet coefficients → SIFT → fused microstructure descriptor`
In the examination, draw each block as a rectangle and connect the blocks with arrows. Under each block write its main intermediate output.

2. Detailed explanation of every stage

1. Scale-space
Create several smoothed/downsampled representations. Large microstructures can be represented at coarse scales while fine structures remain available at finer scales.
Output: Multi-resolution images.
2. Gaussian derivatives
First/second derivatives capture local orientation and intensity curvature while smoothing controls noise amplification.
Output: Derivative maps.
3. Log-Gabor
Use orientation- and frequency-selective filters to represent textured and directional microstructures. Log-Gabor avoids the conventional strong DC component of ordinary Gabor designs.
Output: Frequency/orientation maps.
4. Curvelet
Curvelets provide multi-scale, multi-directional and anisotropic representation, making them suitable for elongated or curved microstructures.
Output: Curvelet coefficients.
5. SIFT
Local keypoints and gradient histograms add distinctive scale/rotation-tolerant local information.
Output: Local descriptors.
6. Fusion
Combine the complementary representations for classification, retrieval or segmentation.
Output: Hybrid feature vector.

3. Mathematical / basic principle

Principle and equations: Scale-space:
$L(x,y,\sigma)=G\sigma * I$

Gaussian derivative:
$\partial L/\partial x$, $\partial L/\partial y$

Gabor/Log-Gabor:
localized orientation/frequency response.

DWT:
LL/LH/HL/HH; Curvelet adds multiple scales and directions.

SIFT:
$m=\sqrt{(Ix^2+Iy^2)}$, $\theta=\text{atan2}(Iy,Ix)$

4. Comparison of alternatives

Technique	Strength	Limitation / trade-off
Selected hybrid	Complementary cues	More stages/parameters
Single detector	Simple and fast	Cannot represent all required information
Alternative robust method	May improve one property	May increase computation

5. Why the selected pipeline is justified

Justification: Canny primarily describes sharp boundaries; Harris emphasizes corners; Hough emphasizes predefined geometric primitives. SEM/TEM microstructures can be elongated, curved, textured and present at multiple scales, so these single methods are incomplete. The hybrid combines scale, direction, frequency, curvature and local distinctiveness. The cost is substantially higher computation and feature dimensionality.

The central design principle is complementarity. A good pipeline assigns a separate role to each stage: preprocessing improves input quality; detection or transformation extracts a meaningful representation; the next stage interprets that representation; and the final stage produces an application-level decision. In segmentation, regional and boundary evidence are combined. In anomaly detection, unsupervised structure is followed by an explicit decision rule. In tracking, spatial foreground detection is combined with temporal displacement.

6. Advantages

- Excellent multi-scale representation.
- Strong orientation and frequency sensitivity.
- Curvelet is suited to anisotropic/curved structures.
- SIFT adds distinctive local matching information.
- The pipeline is modular and can be tuned stage by stage.
- Intermediate outputs are interpretable and can be inspected for errors.
- Complementary cues reduce dependence on a single unreliable image property.

7. Limitations

- Multiple filters/transforms are computationally expensive.
- Feature fusion can produce high-dimensional vectors.
- Parameter selection is important.
- Noise can still generate false local descriptors.
- Parameter selection remains application dependent.
- More stages generally increase computation and memory compared with a single basic operator.

8. Exam-style conclusion

Conclusion: The hybrid strategy is justified because materials microstructures are heterogeneous and anisotropic. A multi-scale, multi-orientation representation is therefore more appropriate than relying on a single edge, corner or line detector.